

Main fixed effects: Ano, Ambiente

Random effect: plot

Nested effect: coleta

Glmer function with link poisson for count data

To set up a hierarchical ANOVA with main effects, nested effects, and random factors in R, the best tool is the `lme4` package. Using the `lmer()` function ensures correct calculations for mixed-effects models, unbalanced designs, and variance components.

[\[1, 2, 3, 4\]](#)

Step-by-Step R Script

R

```
# 1. Load Required Libraries
library(lme4)      # For fitting the mixed-effects model
library(car)       # For Type III Wald chi-square tests (ANOVA table)

# 2. Example Data Setup
# Problem: Yield across different 'Treatments' (Main Fixed Effect)
# and 'Sites' (Random Factor), with 'Plots' nested inside 'Sites'.
set.seed(42)
data <- data.frame(
  Treatment = factor(rep(c("A", "B"), each = 24)),
  Site      = factor(rep(rep(c("Site1", "Site2", "Site3"), each = 8), 2)),
  Plot      = factor(rep(rep(c("P1", "P2", "P3", "P4"), each = 2), 6)),
  Yield     = rnorm(48, mean = 50, sd = 5)
)

# CRITICAL STEP: Create a uniquely coded nested variable.
# While R can handle implicit nesting (Site:Plot), explicit unique coding
# (e.g., Site1_P1, Site2_P1) prevents software misinterpretation.
data$Unique_Plot <- factor(paste(data$Site, data$Plot, sep = "_"))

# 3. Fit the Hierarchical Model
# Structure syntax:
# Fixed Main Effect: Treatment
# Random Factor: (1 | Site)
```

```
# Random Nested Effect: (1 | Site:Plot) OR (1 | Unique_Plot)
model <- lmer(Yield ~ Treatment + (1 | Site) + (1 | Unique_Plot), data =
data)
```

```
# 4. Generate the ANOVA Table
# Use Type III tests if your data has unequal sample sizes per group
anova_table <- Anova(model, type = "III")
print(anova_table)
```

```
# 5. Extract Random Effects (Variance Components)
# This reveals how much variation is added by Site and Plot levels
summary(model)
```

RESULTS NUMBER SEEDS POISSON TRANSFORMATION

```
model <- glmer(Viaveis ~ ambientefactor + anofactor + (1 | Parcela) + (1 |
uniq_coleta), data = consolidado, family = poisson(link = "log"))
```

```
> # Fixed Main Effect: Ano Ambiente
```

```
> # Random Factor: (1 | Parcela)
```

```
> # Random Nested Effect: (1 | Site:Plot) OR (1 | uniq_colet)
```

```
> model <- glmer(Viaveis ~ ambientefactor + anofactor + (1 | Parcela) + (1
| uniq_coleta), data = consolidado, family = poisson(link = "log"))
```

```
boundary (singular) fit: see help('isSingular')
```

```
> # 4. Generate the ANOVA Table
```

```
> # Use Type III tests if your data has unequal sample sizes per group
```

```
> anova_table <- anova(model, type = "III")
```

```
Warning message:
```

```
In anova.merMod(model, type = "III") : additional arguments ignored:
`type`
```

```
> print(anova_table)
```

```
Analysis of Variance Table
```

```
      npar Sum Sq Mean Sq F value
```

```

ambiente factor      3 136245    45415 45415.0

ano factor          2   3608     1804 1804.1

> # 5. Extract Random Effects (Variance Components)

> # This reveals how much variation is added by Site and Plot levels

> summary(model)

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]

Family: poisson ( log )

Formula:

Viaveis ~ ambiente factor + ano factor + (1 | Parcela) + (1 | uniq_coleta)

Data: consolidado

```

AIC	BIC	logLik	-2*log(L)	df.resid
100128.1	100162.3	-50056.0	100112.1	520

Scaled residuals:

Min	1Q	Median	3Q	Max
-34.773	-6.590	-2.468	2.551	102.385

Random effects:

Groups	Name	Variance	Std.Dev.
uniq_coleta	(Intercept)	0.8411	0.9171
Parcela	(Intercept)	0.0000	0.0000

Number of obs: 528, groups: uniq_coleta, 12; Parcela, 4

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.331022	0.264902	16.350	< 2e-16 ***
ambientefactor2	2.250434	0.008606	261.483	< 2e-16 ***
ambientefactor3	-0.010102	0.011606	-0.870	0.384
ambientefactor4	-2.984593	0.046692	-63.921	< 2e-16 ***
anofactor2004	-0.022996	0.005633	-4.082	4.46e-05 ***
anofactor2005	-0.338928	0.006138	-55.219	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)	ambnt2	ambnt3	ambnt4	an2004	
ambintfctr2	-0.029				
ambintfctr3	-0.022	0.671			
ambintfctr4	-0.005	0.167	0.124		
anofctr2004	-0.011	0.000	0.000	-0.035	
anofctr2005	-0.010	0.000	0.000	-0.032	0.455

optimizer (Nelder_Mead) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

To model a percentage or proportion in `glmer` (from the `lme4` package), you should use a **binomial distribution with a logit link function**. The model bounds predictions between 0 and 1, ensuring the probability and log-odds scale correctly. [\[1, 2, 3, 4, 5\]](#)

The two best ways to write the formula depend on how your response variable is structured:

1. If your data is a vector of proportions (0 to 1)

If your response column contains decimals (e.g., `0.75` for 75%), you must use the `weights` argument to specify the total sample size or trial number. [\[1, 2\]](#)

R

```
model <- glmer(proportion_col ~ predictor1 + predictor2 + (1 |
  random_effect),

               weights = total_trials,

               family = binomial(link = "logit"),

               data = your_data)
```

Use code with caution.

2. If you have success and failure counts (Recommended)

If possible, it is mathematically more robust to calculate or provide the exact number of "successes" (the count that makes up your percentage) and "failures" in a `cbind()` function. [\[1\]](#)

R

```
model <- glmer(cbind(successes, failures) ~ predictor1 + predictor2 + (1 |
  random_effect),

               family = binomial(link = "logit"),
```

```
data = your_data)
```

Use code with caution.

3. If you have continuous percentages that aren't based on counts

If your percentages are continuous (e.g., based on area, time, or ratios without a clear integer denominator), a standard binomial model is not appropriate. In this case, you should use a Beta Regression model. While `glmer` does not support Beta directly, you can use packages designed for this like `glmmTMB`. [\[1, 2, 3, 4, 5\]](#)

R

```
library(glmmTMB)

model <- glmmTMB(proportion_col ~ predictor1 + predictor2 + (1 |
  random_effect),

  family = beta_family(link = "logit"),

  data = your_data)
```

RESULTS HIERARCHICAL GLM FOR PROPORTIONS CARIOPSIS

```
anova_table <- anova(model, type = "III")
```

Warning message:

In anova.merMod(model, type = "III") : additional arguments ignored:
"type"

```
> print(anova_table)
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Analysis of Variance Table

	npair	Sum Sq	Mean Sq	F value
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> summary(model)
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